

IWAMIZAWA, Japan

WMO: 474130

Lat: 43.2117N

Lon: 141.7858E

Elev: 52

StdP: 100.71

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-13.5	-11.7	-16.5	0.9	-12.4	-14.9	1.0	-10.8	9.6	-0.6	8.8	-1.5	2.8	50	0.548

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.6	29.1	22.7	27.5	21.8	26.1	20.9	24.0	27.4	23.1	26.1	22.2	24.9	4.4	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
22.9	17.7	25.9	22.1	16.9	25.0	21.2	16.0	24.2	72.5	27.5	69.0	26.3	65.7	25.0	27.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.3	8.3	7.3	-17.2	31.8	2.3	1.3	-18.8	32.7	-20.1	33.5	-21.4	34.2	-23.0	35.2		
			-17.5	25.0	2.2	1.1	-19.1	25.8	-20.4	26.5	-21.7	27.1	-23.3	27.9		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	8.1	-5.7	-4.8	-0.6	6.1	12.4	16.7	20.8	21.9	17.7	10.9	3.8	-2.8	(6)
(7)		DBStd	10.17	3.19	3.44	3.35	3.55	3.21	2.95	2.68	2.69	3.19	3.27	4.03	3.45	(7)
(8)		HDD10.0	1984	486	416	328	127	14	1	0	0	0	29	189	396	(8)
(9)		HDD18.3	3980	744	649	586	367	186	66	9	4	49	231	436	654	(9)
(10)		CDD10.0	1291	0	0	0	10	87	201	334	369	231	56	3	0	(10)
(11)		CDD18.3	246	0	0	0	0	1	16	84	114	31	0	0	0	(11)
(12)		CDH23.3	1186	0	0	0	0	19	86	377	581	123	0	0	0	(12)
(13)		CDH26.7	225	0	0	0	0	3	11	68	129	14	0	0	0	(13)
(14)	Wind	WSAvg	3.4	3.2	3.3	3.5	3.7	3.7	3.3	3.2	3.1	3.0	3.3	3.5	3.3	(14)
(15)	Precipitation	PrecAvg	1235	114	83	59	61	86	72	118	151	150	106	110	125	(15)
(16)		PrecMax	1662	193	161	135	148	151	205	255	344	288	235	192	219	(16)
(17)		PrecMin	933	58	32	20	11	40	23	43	47	60	49	55	74	(17)
(18)		PrecStd	189	33	35	24	33	28	34	56	65	65	39	36	36	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.5	6.0	10.8	21.1	25.6	28.1	30.3	31.8	28.6	21.3	16.4	9.5	(19)
(20)			MCWB	2.8	3.6	6.1	12.0	15.6	19.1	23.5	24.6	22.5	15.7	12.5	7.4	(20)
(21)		2%	DB	2.3	3.9	8.2	17.6	22.7	25.6	28.6	29.8	26.2	19.4	14.1	6.6	(21)
(22)			MCWB	0.7	1.6	4.7	10.5	14.4	18.5	22.4	23.6	20.7	15.1	10.9	4.9	(22)
(23)		5%	DB	0.6	2.3	6.4	14.7	20.4	23.9	27.0	28.0	24.7	18.0	12.4	4.3	(23)
(24)			MCWB	-0.7	0.5	3.6	8.9	13.5	17.9	21.8	22.5	19.9	13.8	9.5	2.6	(24)
(25)		10%	DB	-0.7	0.9	4.8	12.4	18.5	22.1	25.5	26.5	23.1	16.6	10.6	2.4	(25)
(26)			MCWB	-1.9	-0.7	2.3	7.7	12.6	17.0	21.0	21.8	19.0	13.1	8.1	0.7	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.9	3.8	7.0	13.3	17.1	21.0	24.7	25.5	23.7	17.7	13.1	7.7	(27)
(28)			MCD B	4.5	5.5	9.2	18.4	22.8	25.5	28.4	30.3	27.0	19.8	15.6	9.2	(28)
(29)		2%	WB	1.0	2.0	5.4	11.4	15.7	19.6	23.6	24.4	22.3	16.0	11.4	5.1	(29)
(30)			MCD B	2.0	3.3	7.8	16.2	20.4	23.7	26.8	28.2	25.0	18.5	13.7	6.5	(30)
(31)		5%	WB	-0.4	0.7	3.7	9.7	14.6	18.6	22.7	23.5	21.2	14.7	9.8	2.7	(31)
(32)			MCD B	0.6	2.3	5.9	14.0	18.8	22.6	25.6	26.7	23.3	17.2	11.9	4.0	(32)
(33)		10%	WB	-1.9	-0.7	2.5	8.2	13.5	17.6	21.8	22.7	19.8	13.5	8.2	0.9	(33)
(34)			MCD B	-0.8	0.8	4.6	11.8	17.5	21.2	24.6	25.5	22.4	16.0	10.4	2.2	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	6.4	7.2	7.2	9.0	9.8	8.9	7.7	7.6	8.8	8.7	6.8	5.8	(35)
(36)			MCD BR	7.5	8.4	9.0	13.1	13.5	11.8	9.9	9.4	9.6	10.1	9.9	8.3	(36)
(37)			MCWBR	6.6	7.1	6.6	7.3	6.9	5.8	4.6	4.3	4.9	6.3	7.6	7.3	(37)
(38)		5% WB	MCD BR	7.5	8.3	8.5	11.9	11.3	9.9	8.0	8.2	7.7	8.8	9.5	8.2	(38)
(39)			MCWBR	6.9	7.4	6.7	7.4	6.4	5.2	4.1	4.1	4.5	6.3	7.9	7.4	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.342	0.394	0.463	0.523	0.538	0.523	0.507	0.473	0.416	0.408	0.390	0.357	(40)	
(41)		taud	1.975	1.828	1.737	1.775	1.843	1.959	2.071	2.182	2.294	2.172	2.111	1.994	(41)	
(42)		Ebn at Noon	738	756	754	748	752	766	774	788	808	750	686	674	(42)	
(43)		Edh at Noon	117	162	202	211	202	181	160	139	115	114	103	105	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.32	2.09	3.29	4.50	5.13	5.08	4.68	4.38	3.66	2.56	1.45	1.06	(44)	
(45)		RadStd	0.11	0.19	0.29	0.46	0.42	0.47	0.52	0.31	0.31	0.15	0.15	0.10	(45)	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (17 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page